

Task 5: Exploratory Data Analysis (EDA) on the Titanic Dataset

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Domain: Data Analytics

Tools Used:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn

Abstract

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand passenger characteristics, identify trends, detect outliers, and explore relationships between different variables. Python libraries such as Pandas, Matplotlib, and Seaborn were used for data preprocessing, statistical analysis, and visualization.

Objective

The main objectives of this project are:

- To understand the structure of the Titanic dataset.
- To perform basic data exploration.
- To identify missing values and clean the dataset.
- To visualize data distributions using different charts.
- To discover relationships between passenger attributes.
- To summarize important insights from the dataset.

Dataset Description

The Titanic dataset contains passenger information collected from the Titanic disaster.

Fields	Description
PassengerId	Unique Passenger ID
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger Class
Name	Passenger Name
Sex	Gender
Age	Passenger Age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Ticket	Ticket Number
Fare	Ticket Fare

Libraries Used

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

These libraries were used for:

- Data manipulation
- Statistical analysis
- Data visualization

Data Exploration

The following functions were used to understand the dataset:

- `head()`
- `shape`
- `info()`
- `describe()`
- `value_counts()`
- `isnull().sum()`
- `duplicated().sum()`

Findings

- The dataset contains both numerical and categorical features.
- Missing values were identified in the Age column.
- Duplicate records were checked before analysis.

Data Cleaning

The following preprocessing steps were performed:

- Filled missing values in the Age column using the median value.
- Checked for duplicate records.
- Verified data types of each feature.

Exploratory Data Analysis

Question 1

Did a passenger's gender determine their survival?

Visualization Used

- Countplot

Observation

- Female passengers had a significantly higher survival rate than male passengers.
- Most male passengers did not survive.
- Gender played an important role in passenger survival.

Question 2

How does survival rate differ across passenger class?

Visualization Used

- Countplot

Observation

- First-class passengers had the highest survival rate.
- Third-class passengers experienced the highest number of fatalities.
- Passenger class strongly influenced survival probability.

Question 3

What is the relationship between Age, Fare, Passenger Class, and Survival?

Visualization Used

- Pairplot

Observation

- Higher ticket fares were generally associated with first-class passengers.
- Age did not show a strong relationship with Fare.
- Survivors were more concentrated among passengers who paid higher fares.

Question 4

How correlated are the numerical features?

Visualization Used

- Correlation Heatmap

Observation

- Fare has a positive correlation with survival.
- Passenger Class has a negative correlation with Fare.
- Age has only a weak correlation with survival.
- No extremely strong correlations exist among the numerical variables.

Question 5

How is Fare distributed, and what is its relationship with Age?

Visualizations Used

- Boxplot
- Scatter Plot

Observation

Boxplot

- Ticket Fare contains several high-value outliers.
- Survivors generally paid higher fares.

Scatter Plot

- No strong linear relationship exists between Age and Fare.
- Passengers of all ages paid varying ticket fares.
- A few passengers paid exceptionally high fares.

Relationships and Trends Identified

The analysis revealed the following trends:

- Female passengers were more likely to survive.
- First-class passengers had higher survival rates.
- Higher ticket fares were associated with increased survival.
- Passenger class influenced ticket fare significantly.
- Age had only a weak relationship with survival.
- Fare contained several outliers.
- Family-related features showed moderate relationships with each other.

Key Insights

- Most passengers belonged to Third Class.
- Female passengers had better survival chances.

- Higher-paying passengers were more likely to survive.
- Fare distribution is highly right-skewed.
- Passenger Class and Fare are closely related.
- Age alone was not a strong predictor of survival.

Conclusion

This Exploratory Data Analysis successfully examined the Titanic dataset using Python. Various statistical techniques and visualizations were used to understand data distributions, identify outliers, and analyze relationships among features. The analysis highlighted that gender, passenger class, and ticket fare were the most influential factors associated with passenger survival. The project demonstrates how EDA can uncover meaningful insights before building predictive models.